BUSS386 Problem Set 8 — Solutions

Trading Strategies Involving Options

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Problem 1 — Synthetic positions and a parity check

$Ke^{-rT} = 50\,e^{-0.02} = 50 \times 0.98020 = 49.010$.

(a) Parity: $c + Ke^{-rT} = p + S_0$, so $p = c - S_0 + Ke^{-rT} = 4.20 - 51 + 49.010 = \boxed{\$2.21}$.

(b) The market put ($2.80) is $0.59 too expensive. Sell what is dear and buy the replica:

Leg	Cash today	Payoff if $S_T < 50$	Payoff if $S_T \geq 50$
Sell the put	$+2.80$	$-(50 - S_T)$	0
Buy the call	-4.20	0	$S_T - 50$
Short the stock	$+51.00$	$-S_T$	$-S_T$
Lend 49.010	-49.01	$+50$	$+50$
Total	$\boxed{+0.59}$	0	0

You pocket $0.59 today and owe nothing at T in either state.

(c) Synthetic short = short call + long put + borrow Ke^{-rT}. Cash today at market prices: $+4.20 - 2.80 + 49.010 = \boxed{+\$50.41}$. Payoff at T: $-\max(S_T - 50, 0) + \max(50 - S_T, 0) - 50 = -S_T$ in both states, exactly a short share. (At parity prices the cash received would be $51.00; the overpriced put costs you $0.59 of it.)

(d) When actual shorting is banned, restricted or expensive: KRX banned stock short-selling from November 2023 to March 2025, and hard-to-borrow stocks carry fees and recall risk that the option route avoids.

Problem 2 — Principal-protected note (ELB)

(a) Bond cost $10{,}000{,}000\,e^{-0.03} = \boxed{₩\,9{,}704{,}455}$; leftover $\boxed{₩\,295{,}545}$.

(b) Participation rate $\alpha = 295{,}545/850{,}000 = \boxed{34.8\%}$.

(c) Index +20%: $10{,}000{,}000 + 0.348 \times 10{,}000{,}000 \times 0.20 = \boxed{₩\,10{,}695{,}400}$, a 6.95% return. Index −20%: the call expires worthless, you receive $\boxed{₩\,10{,}000{,}000}$.

(d) At $r = 1\%$: bond cost $10{,}000{,}000\,e^{-0.01} = 9{,}900{,}498$, leftover 99,502, and participation $99{,}502/850{,}000 = \boxed{11.7\%}$.
The option budget *is* the interest forgone on the principal, so low rates leave little to buy upside with.

(e) By parity, bond + call = index + put. Split the note as $\alpha \times (\text{bond} + \text{call}) + (1-\alpha) \times \text{bond}$: invest ₩3,477,000 in the index and protect it with an at-the-money put on that amount, and put the rest into a zero-coupon bond with face ₩6,523,000. Check of cost: parity gives the put on ₩10m of index as $850{,}000 - 10{,}000{,}000 + 9{,}704{,}455 = 554{,}455$, so

$$3{,}477{,}000 + 0.3477 \times 554{,}455 + 6{,}523{,}000\, e^{-0.03} = 3{,}477{,}000 + 192{,}784 + 6{,}330{,}216 = 10{,}000{,}000.$$

Problem 3 — The ELS as bond plus short put

(a) Coupons total $3 \times 8 = 24$. Per ₩100:

I_T/I_0	Barrier breached?	Payoff
0.40	yes	40
0.60	yes	60
0.65	no	124
0.80	no	124
1.20	no	124

(b) A put on the index struck at the barrier: the investor has sold crash insurance, and the extra 5% a year is the insurance premium paid out as "coupon."

(c) Bond paying 124 minus a put struck at 0.65 would pay $124 - (65 - 100\, I_T/I_0)$ below the barrier, i.e. 119 at $I_T/I_0 = 0.60$, not 60. The difference, 59, is the same at every level below the barrier. So

$$\text{ELS} = \underbrace{124}_{\text{bond}} - \underbrace{\max(65 - 100\, I_T/I_0,\, 0)}_{\text{short put, strike 0.65}} - \underbrace{59 \times \mathbf{1}\{I_T/I_0 < 0.65\}}_{\text{short cash-or-nothing put}}.$$

The last term is the cliff: at the barrier the 24 of coupons and the 35 of principal protection between 0.65 and 1.00 vanish together. Check at 0.40: $124 - 25 - 59 = 40$.

(d) The real barrier is observed daily (knock-in), so the put is activated if the index *ever* trades below 65%, which is far more likely than finishing below it at maturity; autocall features add further path dependence.

Problem 4 — Covered call on a KOSPI 200 ETF

(a) Premium = $1{,}000 \times$ ₩500 = $\boxed{\text{₩}500{,}000}$.

(b) Portfolio value at T = shares + short-call payoff + premium:

S_T (₩)	Shares	Short call	Total
38,000	38,000,000	0	38,500,000
40,000	40,000,000	0	40,500,000
41,000	41,000,000	0	41,500,000
42,000	42,000,000	−1,000,000	41,500,000
45,000	45,000,000	−4,000,000	41,500,000

(c) Maximum $\boxed{\text{₩}41{,}500{,}000}$, reached for every $S_T \geq 41{,}000$.

(d) The premium absorbs a fall to $\boxed{₩\,39{,}500}$ per share.

(e) Parity: $S_0 - c_0 = Ke^{-rT} - p_0$. Ignoring interest, a covered call is a **bond paying ₩ 41,000 per share plus a short ₩ 41,000 put**, with the ₩ 500 premium kept either way. At $S_T = 38{,}000$: $41{,}000 - (41{,}000 - 38{,}000) + 500 = 38{,}500$ per share, matching the table.

(f) Fund value is capped at ₩ 41,500,000; a plain index ETF is worth ₩ 48,000,000. The covered-call fund lags by $\boxed{₩\,6{,}500{,}000}$, about 13.5%. The "dividend" was the price of that upside.

Problem 5 — Collar, or forward?

(a) Put premium paid \$3.50 = call premium received \$3.50: $\boxed{\text{zero net cost}}$.

(b) Collared value per share = $S_T + \max(70 - S_T, 0) - \max(S_T - 95, 0)$:

S_T	Per share	Total (\$)
60	70	70,000,000
70	70	70,000,000
80	80	80,000,000
95	95	95,000,000
110	95	95,000,000

(c) Minimum $\boxed{\$70{,}000{,}000}$ (floor at K_p), maximum $\boxed{\$95{,}000{,}000}$ (cap at K_c).

(d) Forward price $F_0 = 80\,e^{0.04} = \boxed{\$83.27}$. Shares plus a short forward are worth $S_T + (F_0 - S_T) = F_0$ per share whatever happens: $\boxed{\$83{,}265{,}000}$ at every S_T in the list.

(e) Collar vs. selling: no sale means no capital-gains tax, no loss of voting control, and no signal to the market, yet the downside is floored. Collar vs. forward: the forward is a symmetric hedge that fixes the value at \$83.27 and gives up every outcome above it; the collar is insurance that keeps the outcomes between 70 and 95, at zero premium because the upside above 95 pays for the floor.

Problem 6 — Bull and bear spreads: the mirror rule

(a) Long 45 call, short 55 call. Cash $-8 + 3 = \boxed{-\$5}$ (debit). Payoff 0 / $S_T - 45$ / 10. Max gain $10 - 5 = \boxed{\$5}$ at $S_T \geq 55$; max loss $\boxed{\$5}$ at $S_T \leq 45$; breakeven $\boxed{\$50}$.

(b) Long 45 put, short 55 put. Cash $-3 + 8 = \boxed{+\$5}$ (credit). Payoff -10 / $-(55 - S_T)$ / 0. Profit = payoff $+5$: max gain $\boxed{\$5}$ at $S_T \geq 55$; max loss $\boxed{\$5}$ at $S_T \leq 45$; breakeven $\boxed{\$50}$. Identical profit diagram to (a); only the sign of today's cash differs.

(c) Short 45 call ($+8$), long 55 call (-3): credit $+\$5$.

S_T	Short 45 call	Long 55 call	Payoff	Profit
40	0	0	0	+5
45	0	0	0	+5
50	−5	0	−5	0
55	−10	0	−10	−5
60	−15	+5	−10	−5

Max profit \$5 ($S_T \le 45$), max loss \$5 ($S_T \ge 55$), breakeven \$50.

(d) Bull = buy the low strike and sell the high strike; bear = buy the high strike and sell the low strike; the rule is the same for calls and puts.

(e) Timing of cash and margin: the credit version holds the premium (and earns interest on it) while the debit version pays up front; with American options the in-the-money short leg of a credit spread can be assigned early. The diagrams are the same by parity; the frictions are not.

Problem 7 — Butterfly and the convexity check

(a) Long 90 call, long 110 call, short two 100 calls. Cost $13 + 3 - 2 \times 7 =$ \boxed{\$2}.

(b)

S_T	$(S_T - 90)^+$	$(S_T - 110)^+$	$-2(S_T - 100)^+$	Payoff
85	0	0	0	0
95	5	0	0	5
100	10	0	0	10
105	15	0	−10	5
115	25	5	−30	0

(c) The payoff is never negative and is positive between 90 and 110, so a zero or negative price would be a free lunch. Equivalently $c(100) \le \frac{1}{2}\big(c(90) + c(110)\big)$: call prices are convex in the strike. Here $7 \le 8$: \boxed{\text{no arbitrage}}.

(d) Cost becomes $13+3-2\times 8.5 = -\$1$: you are *paid* \$1 to hold a non-negative payoff. \boxed{\text{Arbitrage}}: buy the 90 and 110 calls, sell two 100 calls, keep the \$1, and collect up to \$10 more at expiry. Convexity is violated ($8.5 > 8$).

(e) Two naked short calls bring in \$14 but lose $2(S_T - 100) - 14$ for $S_T > 107$, without limit. The butterfly adds the two wings for \$16, turning the \$14 credit into a \$2 debit, and caps the loss at \boxed{\$2}. The \$16 is the price of the cap.

Problem 8 — Calendar spread

(a) $+7.76 - 11.35 =$ \boxed{-\$3.59} (debit).

(b) Short call settles at $-(130 - 100) = -30$. The remaining call has six months to run, so by Lecture 7

$$\max\big(130 - 100\,e^{-0.015},\, 0\big) = 130 - 98.511 = 31.489 \;\le\; c \;\le\; 130.$$

Profit $= -30 + c - 3.59 \in \boxed{[-2.10,\ 96.41]}$. With the stock far above the strike the long call is almost all intrinsic value, and the realistic outcome is near the lower end: the "calm market" bet failed.

(c) Short call settles at 0. Bounds:

$$\max(100 - 98.511,\, 0) = 1.489 \ \le\ c\ \le\ 100.$$

Profit $\in \boxed{[-2.10,\ 96.41]}$ again. For any $S_{T_1} \ge K$ the lower bound is $K(1 - e^{-r\tau}) - 3.59$, the same number. The bounds cannot tell us whether the trade made money: that depends on the market price of a live option.

(d) If a 6-month at-the-money call is worth what it was today, $c = 7.76$ and profit $= 7.76 - 3.59 = \boxed{+\$4.17}$. This is the peak of the calendar's profit curve in the lecture.

(e) The calendar bets on how the market will *value* the remaining option at T_1, i.e. on implied volatility (that near-term volatility deflates relative to long-term); the butterfly's payoff is fixed by S_T alone and needs no model.

Problem 9 — Straddle vs. strangle

(a) Straddle $4 + 4 = \boxed{\$8}$. Strangle $2 + 2 = \boxed{\$4}$.

(b) Straddle: $100 \pm 8 = \{92, 108\}$. Strangle: $95 - 4 = 91$ and $105 + 4 = 109$.

(c)

S_T	Straddle profit	Strangle profit
85	$15 - 8 = +7$	$10 - 4 = +6$
92	$8 - 8 = 0$	$3 - 4 = -1$
100	$0 - 8 = -8$	$0 - 4 = -4$
108	$8 - 8 = 0$	$3 - 4 = -1$
115	$15 - 8 = +7$	$10 - 4 = +6$

(d) Small move: both lose, the strangle less (-4 vs. -8 at $S_T = 100$) because it cost less. Large move: both win, the straddle by \$1 more at every S_T beyond the strikes because its payoff starts growing at 100 rather than at 95/105. Cheaper entry, wider dead zone.

Problem 10 — Iron condor on SPX 0DTE

(a) Long 5,750 put, short 5,780 put, short 5,820 call, long 5,850 call. That is a **bull put spread** (credit) plus a **bear call spread** (credit): a short strangle with both wings bought back.

(b) Max profit = credit $= 8 \times 100 = \boxed{\$800}$, for $5{,}780 \le S_T \le 5{,}820$. Max loss = wing width minus credit $= (30 - 8) \times 100 = \boxed{\$2{,}200}$, for $S_T \le 5{,}750$ or $S_T \ge 5{,}850$. Breakevens 5,772 and 5,828.

(c) The market's implied daily move (which sets the \$8 credit) must on average exceed the realized move, so that the index finishes inside the short strikes often enough: ignoring partial losses, the trader needs to win more than $2{,}200/(800 + 2{,}200) \approx 73\%$ of the time. The strategy harvests the volatility risk premium and pays for it on tail days.

Problem 11 — Box spread financing

(a) Long K_1 call, short K_2 call, short K_1 put, long K_2 put. At expiry

$$(S_T - K_1)^+ - (S_T - K_2)^+ + (K_2 - S_T)^+ - (K_1 - S_T)^+ = K_2 - K_1 = \boxed{\$200}$$

in each of the three regions $S_T < K_1$, $K_1 \le S_T < K_2$, $S_T \ge K_2$.

(b) $197.50 = 200\, e^{-r^* \times 0.5}$ gives

$$r^* = -2 \ln\left(\frac{197.50}{200}\right) = -2 \ln 0.9875 = \boxed{2.52\%}.$$

(c) $r^* = 2.52\% < 4\%$: the box is $\boxed{\text{rich}}$. Buying it lends at 2.52% when Treasuries pay 4%, so an arbitrageur *sells* the box (borrows at 2.52%) and buys Treasuries.

(d) The box rate reflects the credit of the clearing house and the capital tied up in four option positions rather than Treasury credit, plus early-assignment risk if the options are American and demand imbalances when traders use boxes for cheap leverage.

Problem 12 — Matching the strategy to the view

(i) **Bull spread.** Pay (with calls) or receive (with puts). Mildly bullish, calm.

(ii) **Zero-cost collar**: long put, short call, strikes chosen so the premiums match. Zero net. Protect the downside; view on volatility implicit (you sell the upside you do not expect).

(iii) **Long straddle** (or strangle if cheaper). Pay. No direction, big move.

(iv) **Long butterfly** centred at today's price. Pay a small premium; worst case is that premium. No direction, calm. (A calendar spread expresses the same view but its worst case depends on the remaining option's price.)

(v) **Long box spread**, if its implied rate exceeds the deposit rate. Pay $(K_2 - K_1)e^{-r^*T}$ today, receive $K_2 - K_1$ in six months. No view on the stock at all.

(vi) **Synthetic short**: short call, long put, borrow Ke^{-rT}. Roughly zero net (receive the forward value of the stock). Bearish, any volatility.

(vii) **Covered call.** Receive. Neutral to mildly bullish, calm; you have sold the upside above the strike.